

Winter Internship Training Program
Project Work (Rocketry Training)

Project Title: OPENROCKET
ROCKET DESIGN ASSIGNMENT

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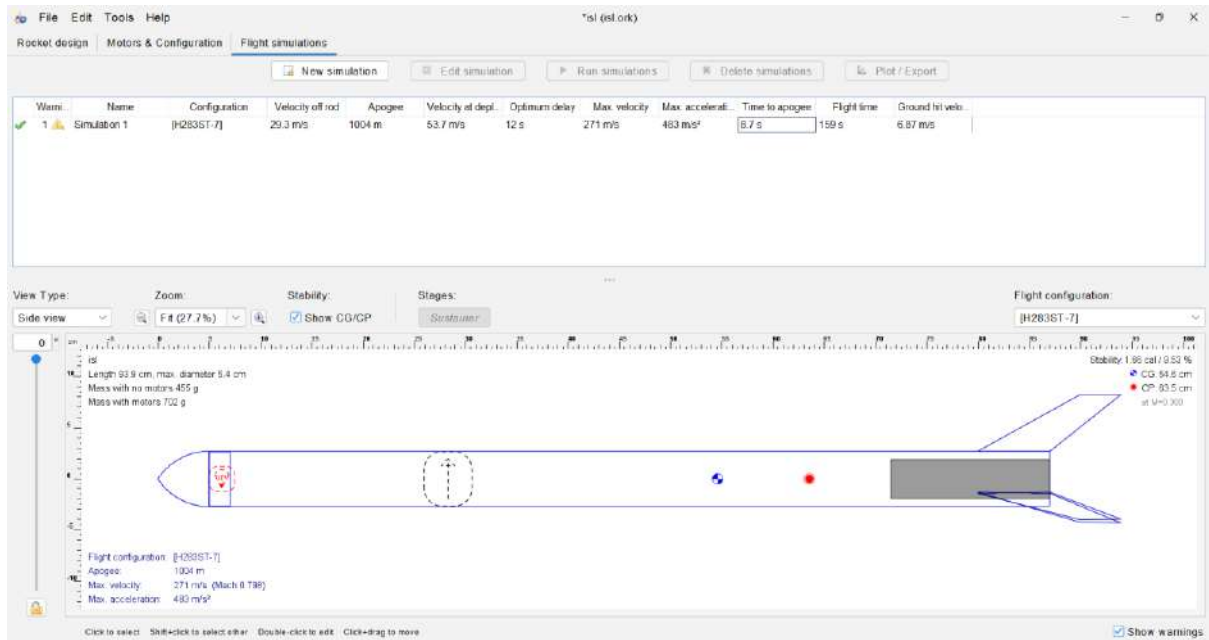
Introduction-:

The objective of this project is to design, simulate, and optimize a single-stage model rocket using OpenRocket.

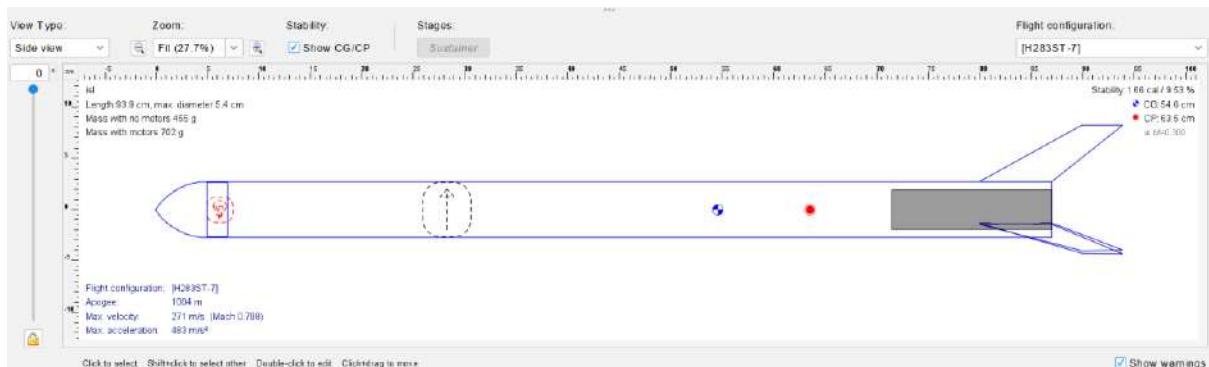
The design must meet all mission requirements and constraints, including apogee ≥ 1000 m, stability margin between 1.0–2.0 calibers, safe recovery (<10 m/s), and subsonic flight (Mach <0.8).

Mission Requirements Verification:-

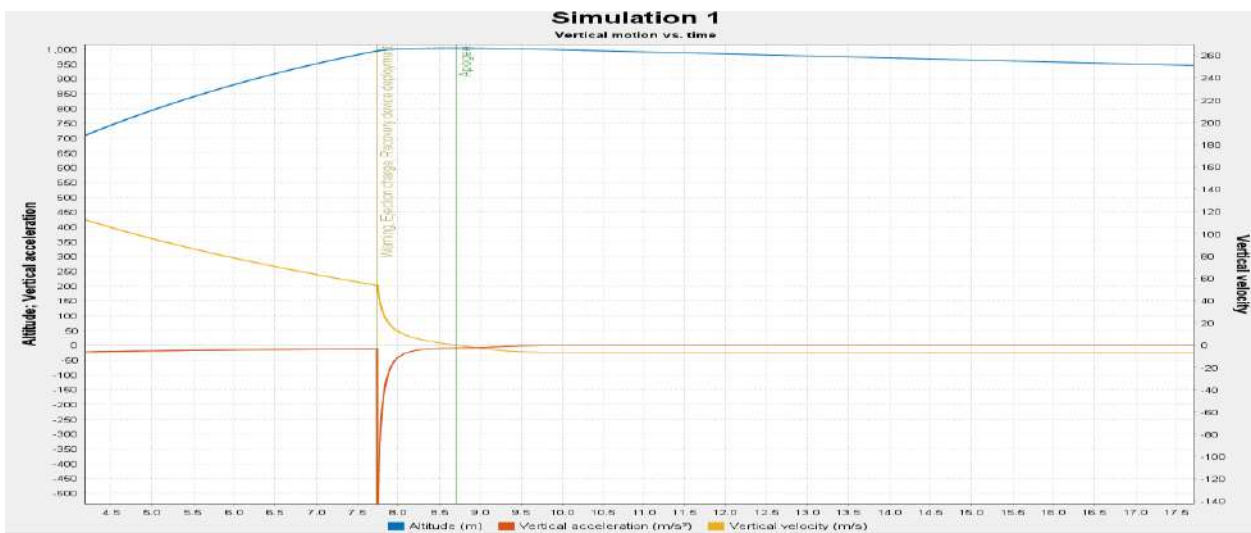
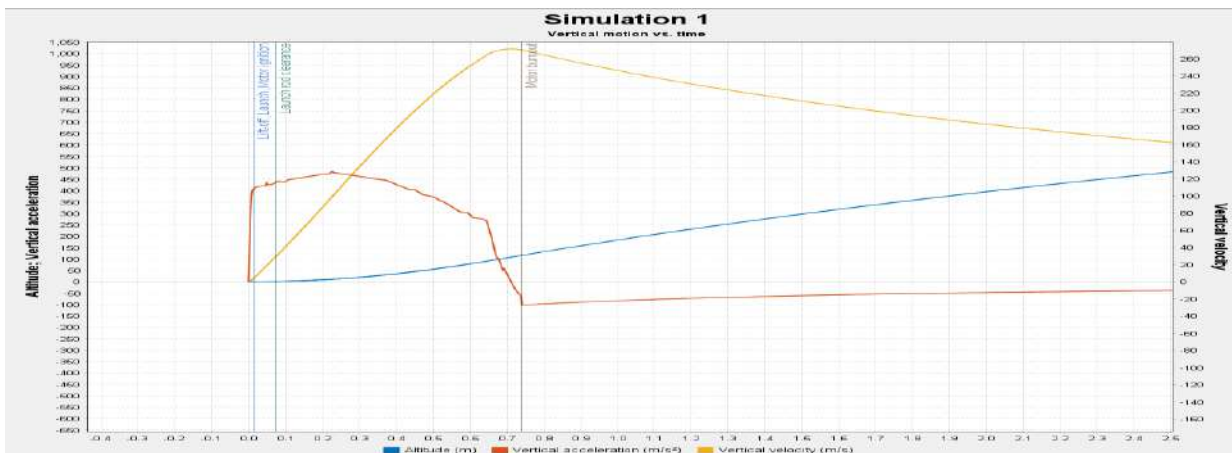
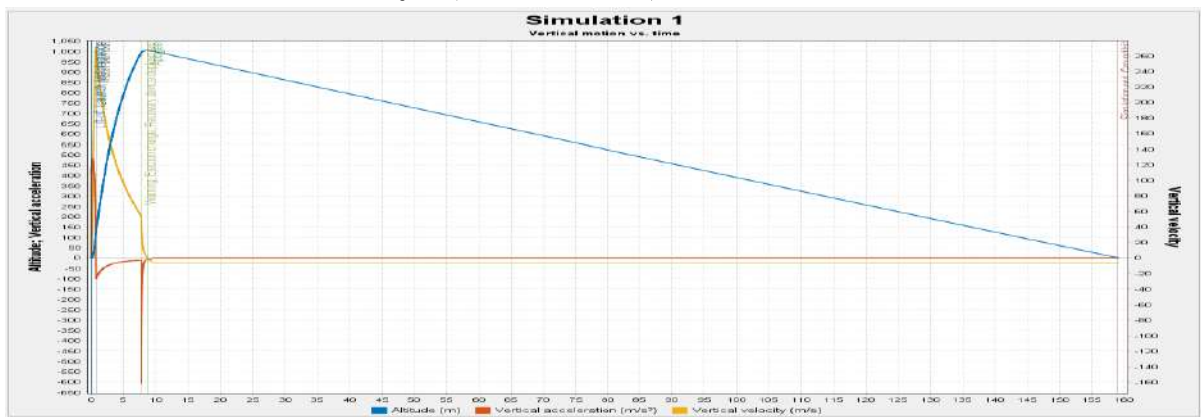
- Apogee ≥ 1000 m $\rightarrow$



- Stability margin 1.0–2.0 cal $\rightarrow$



- Safe recovery (<10 m/s) $\rightarrow$



“Figure : Altitude, velocity, and acceleration vs. time plot showing smooth ascent, apogee at 1004 m, and safe descent rate of 6.87 m/s after parachute deployment.”

“The plot confirms that the rocket reached the required apogee, remained stable, and recovered safely with a descent rate below 10 m/s.” “The rocket consistently recovers at ~ 6.8 m/s, which is below the 10 m/s safety threshold.”

- Single-stage configuration →



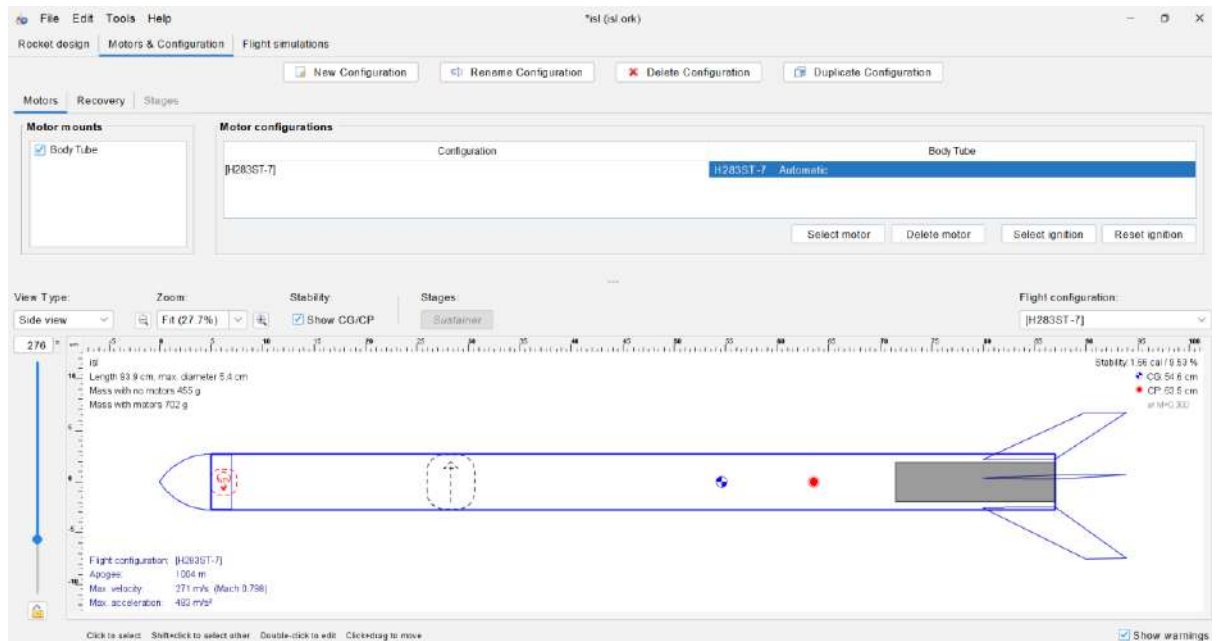
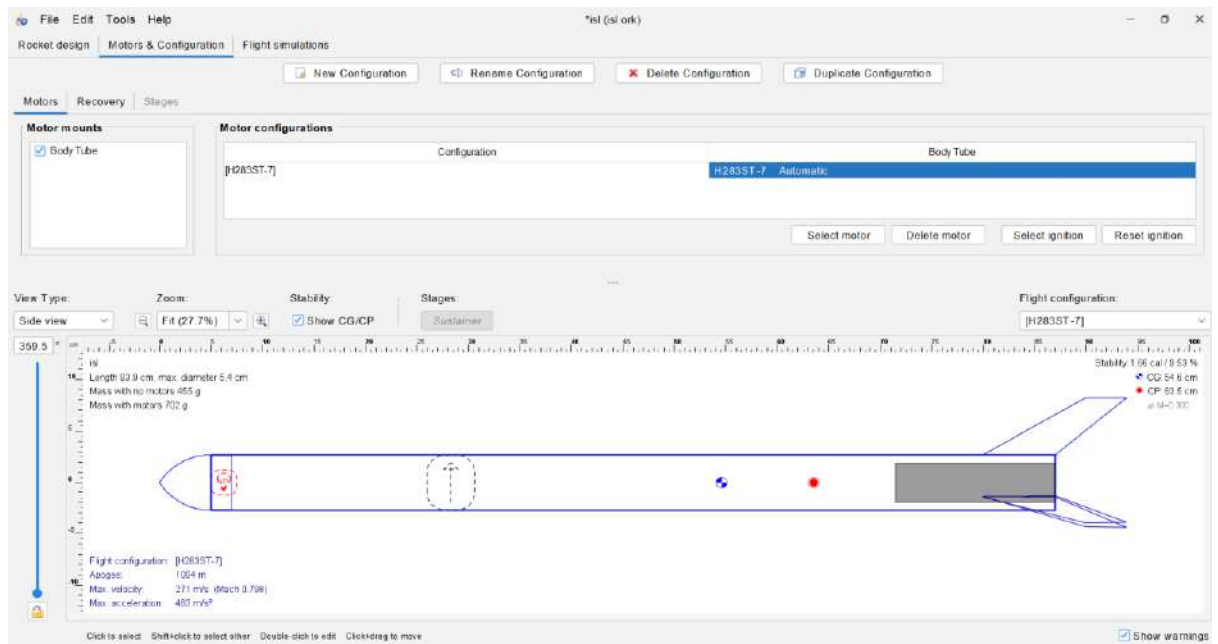
Design Constraints:-

- ❖ **Total Length:** 939 mm (nose cone + body tube + aft section), within the required 800–1200 mm range.
- ❖ **Body Diameter:** 54 mm (fixed).
- ❖ **Lift-off Mass:** 702 g (including motor), well below the 1.5 kg maximum.
- ❖ **Materials:** Fiberglass (body tube and fins), Polystyrene (nose cone), Ripstop Nylon (parachute).
- ❖ **Maximum Mach Number:** 0.798 (<0.8 , subsonic).

Fins Subsystem:-

A trapezoidal fin set with airfoil cross-section was selected to balance aerodynamic efficiency and structural strength.

- ❖ **Configuration:** 3 fins, trapezoidal planform, airfoil profile.
- ❖ **Material:** Fiberglass, chosen for durability and stiffness under high acceleration ($\sim 483 \text{ m/s}^2$).
- ❖ **Aerodynamic Justification**
 - The airfoil profile reduces drag compared to flat fins, improving apogee performance.
 - Three fins provide sufficient stability without excessive drag.
 - The chosen dimensions shift the center of pressure appropriately, resulting in a stability margin of **1.66 calibers**, which lies within the required 1.0–2.0 range.



Subsystem Design:-

❖ **Nose Cone**

- ❖ *Type: Ogive*
- ❖ *Dimensions: Length 5 cm, Base diameter 5.4 cm, Wall thickness 0.2 cm*
- ❖ *Shoulder: Diameter 5.2 cm, Length 2 cm, Thickness 0.2 cm, End-capped*
- ❖ *Material: Polystyrene (density 1.05 g/cm³)*
- ❖ *Mass: 23.3 g*

Justification:

An ogive nose cone was selected for its aerodynamic efficiency, minimizing drag and ensuring smooth airflow over the rocket body. The tangent ogive profile (shape parameter = 1) provides a seamless transition to the body tube, reducing flow separation and improving apogee performance. The 0.2 cm wall thickness balances strength and low mass, keeping the rocket lightweight yet durable.

The shoulder design (5.2 cm diameter, 2 cm length, end-capped) ensures a secure fit into the body tube, preventing air leakage and maintaining structural alignment during flight. The end-cap feature adds rigidity, reducing the risk of deformation under aerodynamic loads. Polystyrene was chosen for its ease of shaping, sufficient rigidity, and compatibility with the fiberglass body tube. Overall, this configuration improves apogee while keeping the center of pressure forward, contributing to the stability margin of ~1.66 calibers.

Nose Cone configuration

Component name:

General | **Shoulder** | **Override** | **Appearance** | **Comment**

Nose cone shape:

Shape parameter:

Length:

Base diameter:

☐ Automatic

Wall thickness:

☐ Filled

☐ Flip to tail cone

Material

Component material:

Component finish:

Component mass: 23.3 g

Nose Cone configuration

Component name:

General | **Shoulder** | **Override** | **Appearance** | **Comment**

Diameter:

Length:

Thickness:

☒ End capped

Component mass: 23.3 g

❖ Fins

- **Type:** Trapezoidal, 3 fins
- **Dimensions:** Root chord 7 cm, Tip chord 4 cm, Height 5.3 cm, Sweep length 9.93 cm, Sweep angle 61°, Thickness 0.3 cm
- **Material:** Fiberglass (density 1.85 g/cm³)
- **Cross-section:** Airfoil

Justification:

*The trapezoidal fin geometry was selected to balance aerodynamic efficiency and structural strength. The **airfoil cross-section** reduces drag compared to flat fins, improving apogee performance while maintaining smooth airflow. The **sweep angle of 61°** contributes to shifting the center of pressure rearward, ensuring a stability margin of ~1.66 calibers within the required 1.0–2.0 range. The **0.3 cm thickness** provides sufficient rigidity to withstand high acceleration ($\sim 483 \text{ m/s}^2$) without deformation. Using **fiberglass** ensures durability and stiffness, allowing the fins to maintain their aerodynamic profile under flight loads. Three fins were chosen to provide adequate stability without introducing excessive drag. Additionally, the inclusion of **through-the-wall fin tabs (5 cm length)** enhances structural integrity by anchoring the fins directly into the body tube. This reinforcement prevents fin flutter and detachment under aerodynamic loads, ensuring reliable stability throughout flight. The use of **fiberglass** further guarantees stiffness and durability, allowing the fins to maintain their aerodynamic profile under repeated launches.*

Trapezoidal Fin Set configuration

Component name: Trapezoidal Fin Set

General Fin tabs Override Appearance Comment

Number of fins: 3

Fin cant: 1°

Root chord: 7 cm

Tip chord: 4 cm

Height: 5.5 cm

Sweep length: 9.93 cm

Sweep angle: 61°

Fin cross section: Airfoil

Thickness: 0.3 cm

Placement

Position relative to: Bottom of the parent component

plus 0 cm

Fin rotation: 0°

Material

Component material: Fiberglass (1.85 g/cm³)

Component finish: Regular paint (60 µm) Set for all

Root Fillets

Fillet radius: 0 cm

Fillet material: Balsa (0.17 g/cm³)

Component mass: 42.8 g Split fins Convert to freeform Export to SVG Cancel OK

Trapezoidal Fin Set configuration

Component name: Trapezoidal Fin Set

General Fin tabs Override Appearance Comment

Through-the-wall fin tabs:

Tab length: 5 cm

Tab height: 0 cm

Tab position: 0 cm

relative to Middle of the parent component

Calculate automatically

Component mass: 42.8 g Split fins Convert to freeform Export to SVG Cancel OK

❖ Propulsion

Motor: H283ST-7 (certified H-class, Super Thunder propellant)

❖ Specifications:

- Total impulse: 199 Ns
- Average thrust: 270 N
- Peak thrust: 337 N
- Burn time: 0.738 s
- Loaded mass: 247 g

❖ Thrust-to-Weight Ratio: ~ 41.1 ($>5:1$ required)

Thrust-to-weight ratio: $41.1(>5:1)$

Justification: Provides sufficient thrust to reach ≥ 1000 m apogee while keeping Mach < 0.8 . The selected H283ST-7 motor produced a thrust-to-weight ratio of $\sim 41:1$, which is significantly higher than the minimum requirement of $5:1$.

Although this ratio indicates a very aggressive initial acceleration (~ 483 m/s²), the design was balanced by careful payload selection and fin geometry.

These adjustments ensured that the rocket remained stable (1.66 calibers) and subsonic (Mach 0.798) throughout flight.

Thus, the high thrust-to-weight ratio was acceptable and beneficial, as it guaranteed reliable lift-off and achievement of the target apogee without compromising safety or stability.

Select a rocket motor

Select thrust curve: H283ST

Ejection charge delay: 7

(or type in desired delay in seconds)

☒ Hide very similar thrust curves

☒ Hide motors which are not in regular production

Motor name column: ☐ Use common name ☒ Use manufacturer's designation

Manufacturer	Name	Total Impulse	Case	Diameter ^	Length
Cesaroni Technology L...	127/G46-11A	124	Pro38-1G	38 mm	127 mm
Cesaroni Technology L...	128G185-1...	127	Pro38-1G	38 mm	127 mm
Cesaroni Technology L...	129G79-13A	130	Pro38-1G	38 mm	125 mm
AeroTech	G69N	136	RMS-38/120	38 mm	106 mm
Loki Research	G66-LR	136	38/120	38 mm	122 mm
Cesaroni Technology L...	137G58-13A	137	Pro38-1G	38 mm	127 mm
Cesaroni Technology L...	134G60-14A	139	Pro38-1G	38 mm	127 mm
Cesaroni Technology L...	141G115-1...	141	Pro38-1G	38 mm	127 mm
Cesaroni Technology L...	141G78-15A	143	Pro38-1G	38 mm	126 mm
Contrail Rockets	G123-HP	143	38 16 S HP ...	38 mm	406 mm
Contrail Rockets	G100-PVC	146	38 16 S PV...	38 mm	406 mm
Cesaroni Technology L...	150G50-15A	150	Pro38-1G	38 mm	127 mm
Contrail Rockets	H222-HP	161	38 16 M HP ...	38 mm	407 mm
Contrail Rockets	H300-HP	163	38 20 F HP ...	38 mm	508 mm
Contrail Rockets	H303-PVC	172	48 20 F PV...	38 mm	508 mm
AeroTech	H73J	188	RMS-38/240	38 mm	152 mm
AeroTech	H283ST	199	DMS	38 mm	155 mm
Loki Research	H100-SF	201	38/240	38 mm	177 mm
Contrail Rockets	H211-PVC	208	38 20 M PV...	38 mm	516 mm
AeroTech	H130W	210	RMS-38/240	38 mm	152 mm
AeroTech	H148R	215	RMS-38/240	38 mm	152 mm
AeroTech	H669N	220	RMS-38/240	38 mm	152 mm

Number of motors: 806

Search:

Filter Motors

Show Details

Designation: H283ST

Common name: H283

Total impulse: 199 Ns (24% H)

Avg. thrust: 270 N

Max. thrust: 337 N

Burn time: 0.738 s

Launch mass: 247 g

Empty mass: 150 g

Motor type: Single-use

Case info: DMS

Propellant: Super Thunder

Compatible Cases:

Data points: 36

38mm DMS Super Thunder

Thrust curve:

OK

Cancel

❖ Payload Subsystem

- **Type:** *Payload + Altimeter*
- **Mass:** *111 g (≥ 75 g required)*
- **Dimensions:** *Length 4.62 cm, Diameter 5.2 cm*
- **Material/Density:** *Approx. 1.13 g/cm³*
- **Placement:** *Axially centered (radial distance 0 cm, radial direction 0°), positioned near the middle of the parent body tube (offset -17.7 cm)*

Justification:

*The payload subsystem includes both a physical payload mass and an **altimeter** for apogee detection. The total mass of 111 g exceeds the minimum requirement of 75 g, ensuring compliance with certification constraints. The cylindrical geometry (4.62 cm length, 5.2 cm diameter) allows secure integration within the body tube without disturbing aerodynamic flow.*

*Precise **radial positioning (0 cm offset, 0° direction)** ensures the payload remains perfectly centered, preventing asymmetry that could destabilize flight. Placement near the rocket's mid-section balances the center of gravity, contributing to the overall stability margin of ~1.66 calibers. The inclusion of the altimeter guarantees accurate apogee detection and reliable deployment of the recovery system*

Mass Component configuration

Component name: Payload + Altimeter

General Radial position Override Appearance Comment

Mass type: Altimeter

Mass: 111 g

Approximate density: 1.13 g/cm³

Length: 4.62 cm

Diameter: 5.2 cm

☒ Automatic

Placement

Position relative to: Middle of the parent component

plus -17.7 cm

Component mass: 111 g

Cancel OK

Mass Component configuration

Component name: Payload + Altimeter

General Radial position Override Appearance Comment

Radial distance: 0 cm

Radial direction: 0 °

Röset

Component mass: 111 g

Cancel OK

❖ Recovery System

- ❖ *Recovery System Subsystem*
- ❖ **System:** Parachute + shock cord/harness
- ❖ **Canopy:** Diameter 60 cm, Material Ripstop Nylon (67 g/m^2), Drag coefficient $C_D=0.800$
- ❖ **Shroud Lines:** 6 lines, Length 60 cm, Material elastic cord (2 mm round, 1.8 g/m)
- ❖ **Packed Dimensions:** Length 2.5 cm, Diameter 2.5 cm
- ❖ **Deployment:** First ejection charge, at ~200 m altitude
- ❖ **Mass:** 25.4 g
- ❖ **Radial Positioning:** Centered (radial distance 0 cm, radial direction 0°)
- ❖ **Descent Rate:** ~6.8 m/s (<10m/s requirement)

Justification:

The recovery system uses a **ripstop nylon parachute** with a 60 cm canopy, chosen for its high strength-to-weight ratio and tear resistance. The **drag coefficient of 0.800** ensures sufficient aerodynamic braking, resulting in a descent rate of ~6.8 m/s, well below the 10 m/s safety threshold. Six

elastic shroud lines distribute load evenly, preventing canopy collapse and ensuring stable descent.

- **Precise radial positioning (0 cm offset, 0° direction)** keeps the parachute perfectly centered, avoiding asymmetry that could destabilize recovery.

The compact packed dimensions (2.5 cm × 2.5 cm) allow efficient integration into the airframe. Deployment at ~200 m altitude ensures timely canopy inflation for safe recovery. The inclusion of a **shock cord/harness** secures airframe components during descent, preventing separation damage. Overall, this subsystem guarantees compliance with training requirements and ensures safe, controlled recovery of the rocket.

The screenshot shows a 'Parachute configuration' dialog box with the following settings:

- Component name:** Parachute
- Custom:** Custom
- Parts Library:** Parts Library
- General Tab:**
 - Canopy:**
 - Diameter: 60 cm
 - Material: Ripstop nylon (67 g/m²)
 - Drag coefficient C_D : 0.800 (Reset button)
 - Shroud lines:**
 - Number of lines: 6
 - Line length: 60 cm
 - Material: Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)
- Placement Tab:**
 - Position relative to: Top of the parent component
 - plus 0 cm
 - Packed length: 2.5 cm
 - Packed diameter: 2.5 cm
 - ☐ Automatic
- Deployment Tab:**
 - Deploys at: + First ejection charge of this stage
 - plus 0 seconds
 - Altitude: + 200 m
 - + This parameter can be overridden in each flight configuration.

At the bottom, the component mass is listed as 25.4 g. There are Cancel and OK buttons.

✖

Parachute configuration

×

Component name:

Parachute

Custom

Parts Library

General

Radial position

Override

Appearance

Comment

Radial distance:

0

cm

Radial direction:

0

°

Reset

i

Component mass: 25.4 g

Cancel

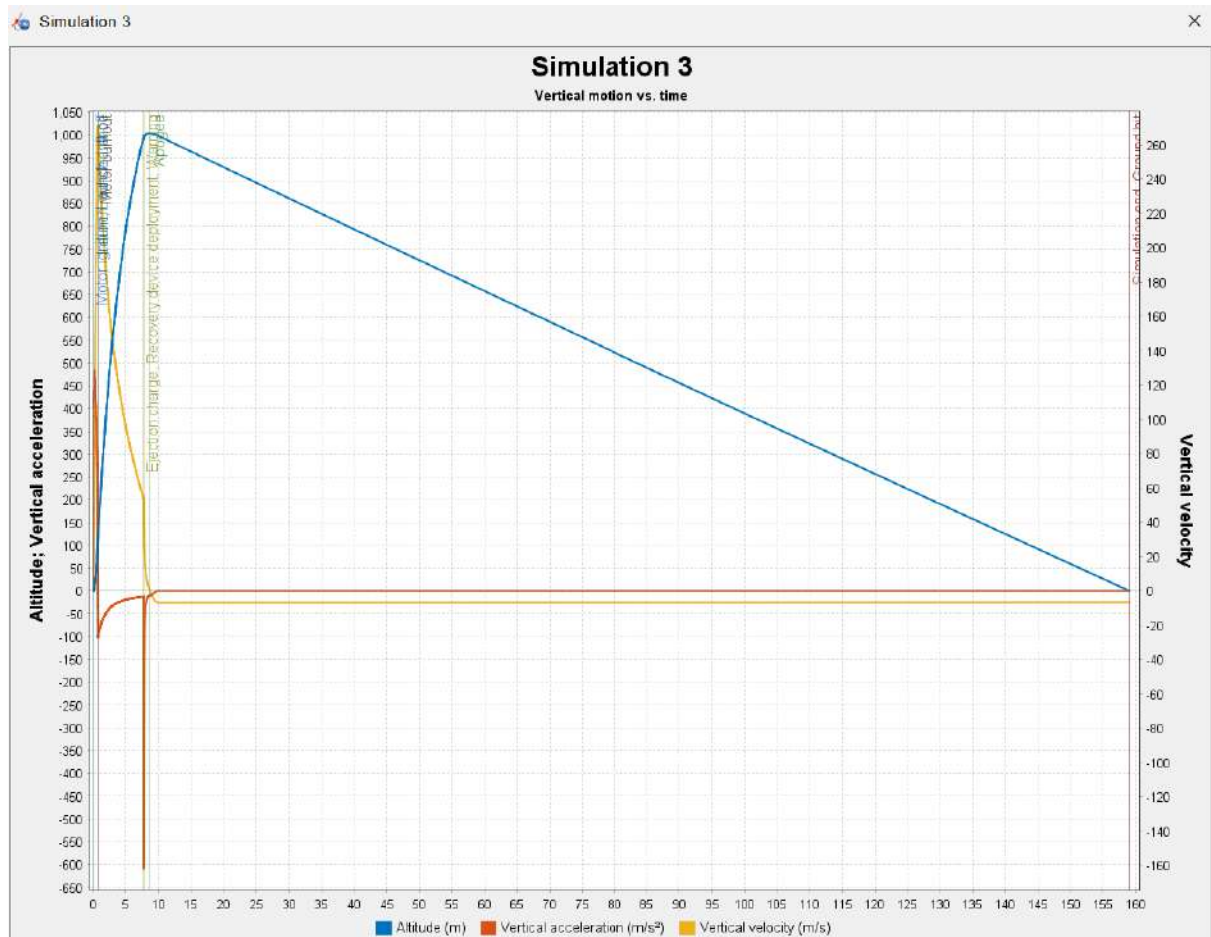
OK

Performance Verification:-

Performance verification was conducted using OpenRocket simulations.

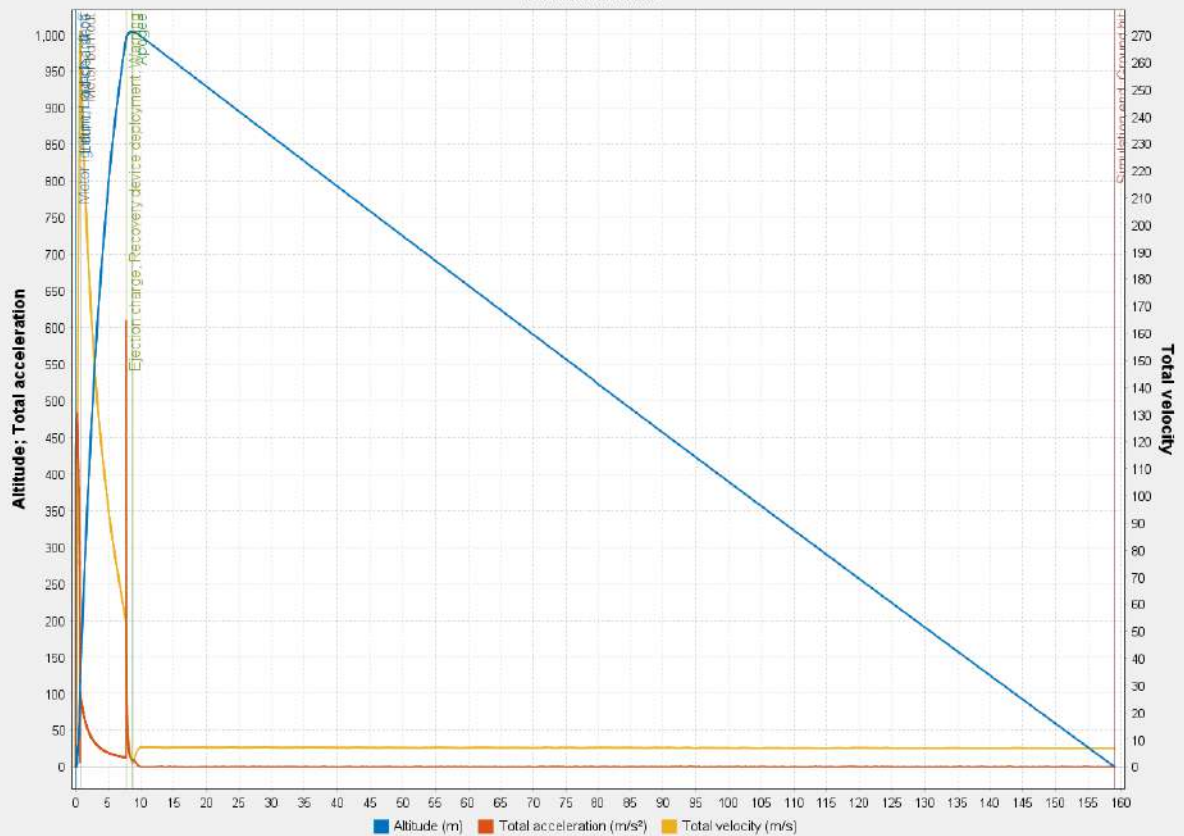
Key parameters were recorded to confirm compliance with design constraints and training requirements.

The following screenshots provide evidence of apogee, stability margin, velocity, acceleration, and aerodynamic balance (CG vs CP).



Simulation 3

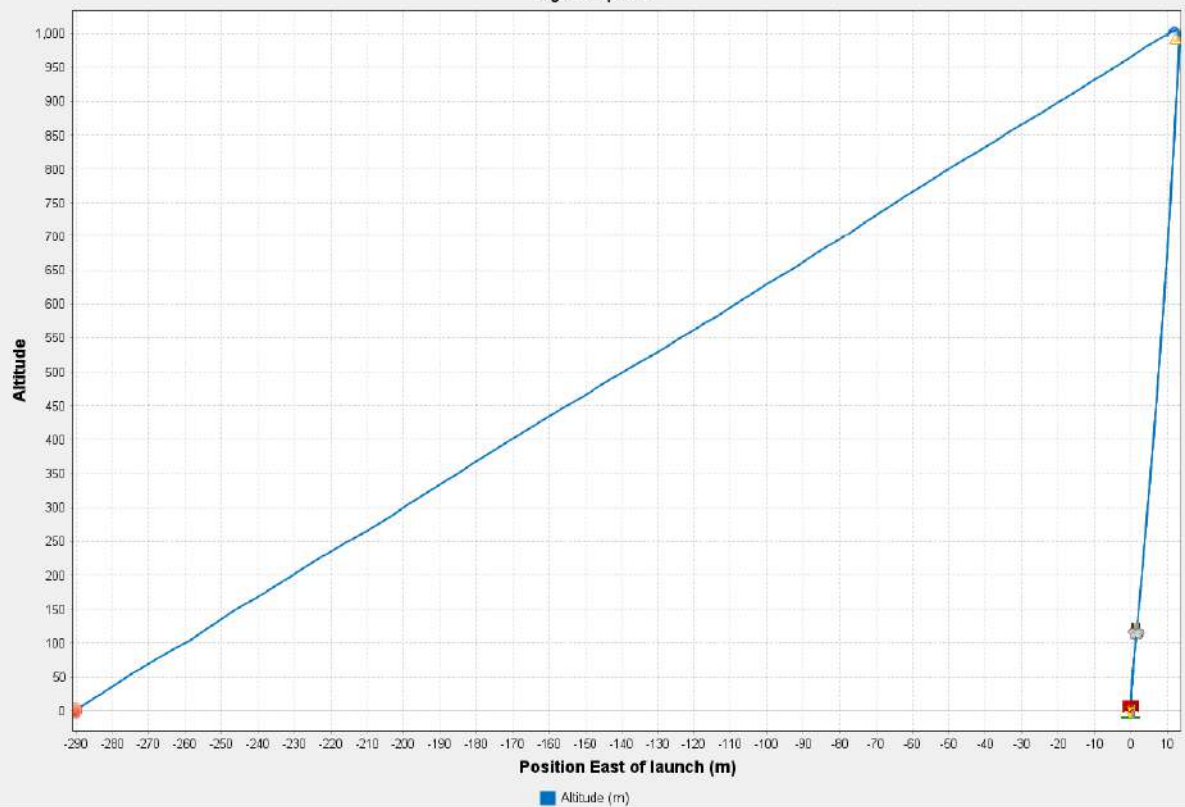
Total motion vs. time



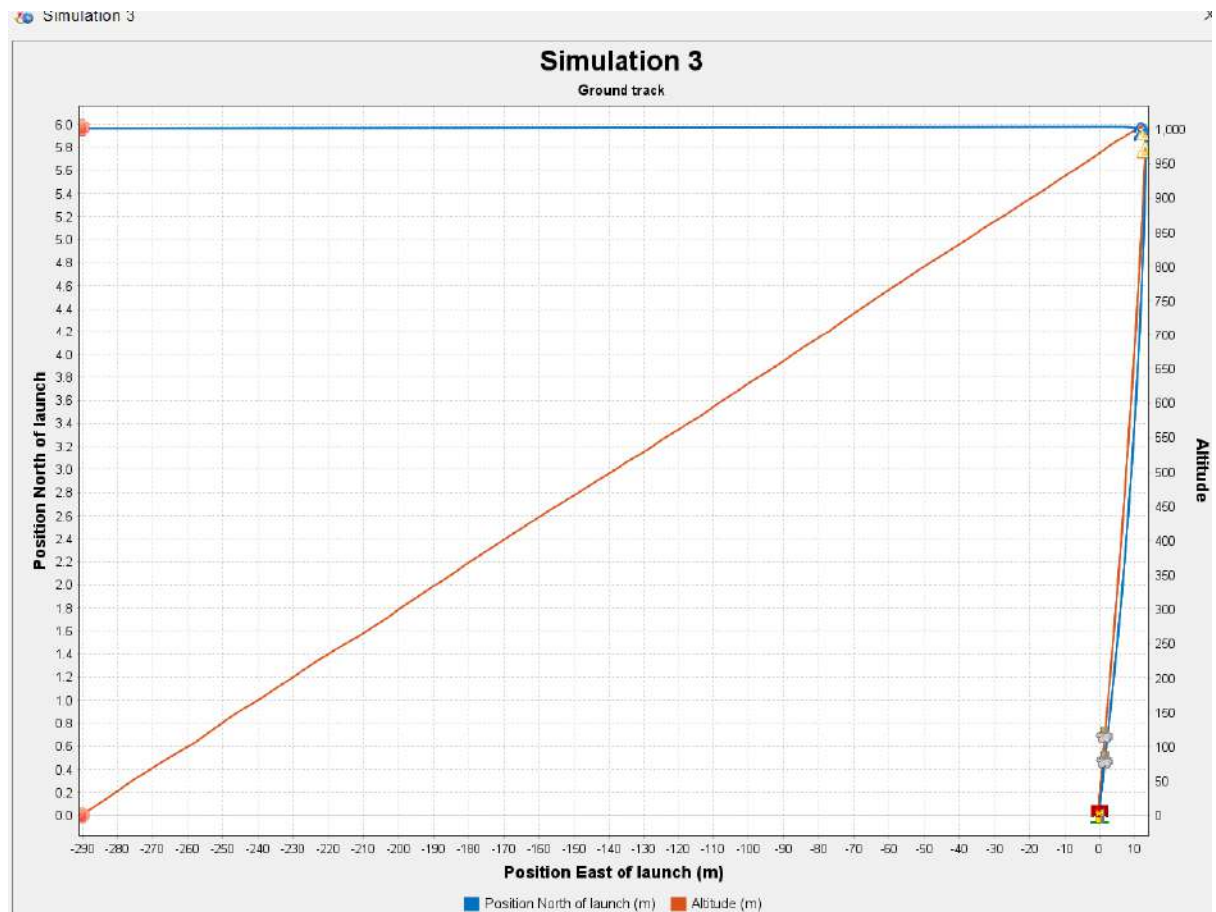
Simulation 3

Simulation 3

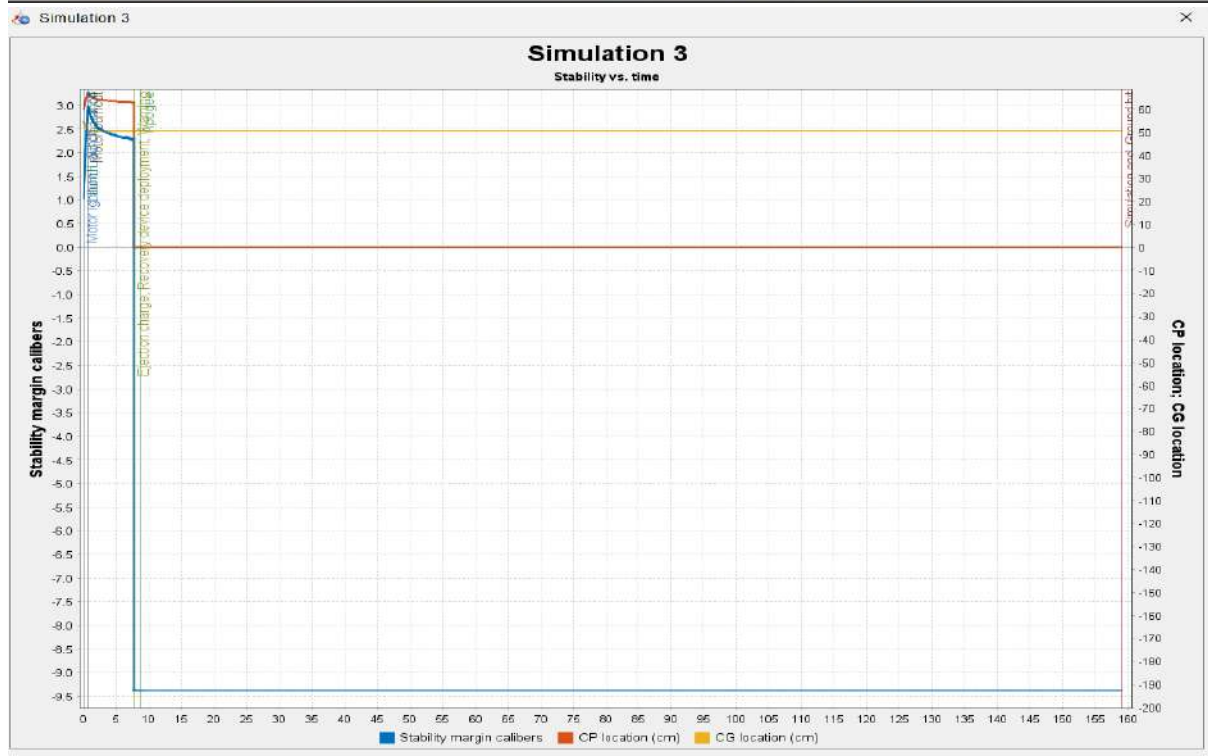
Flight side profile

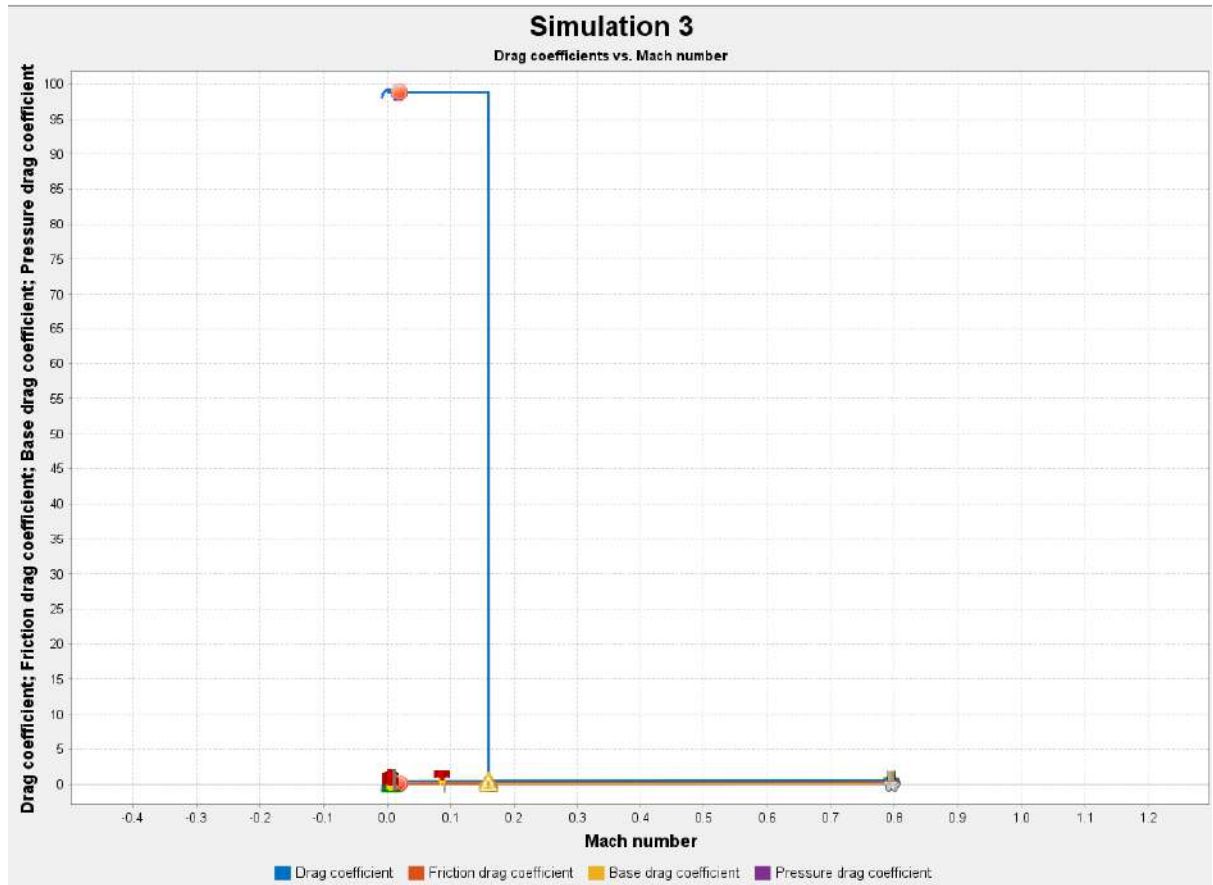


left click drag to zoom area. mouse wheel to zoom. ctrl-mouse wheel to zoom x axis only. ctrl-left click drag to pan. right click drag to zoom dynamically.



left click drag to zoom area, mouse wheel to zoom, ctrl-mouse wheel to zoom x axis only, ctrl-left click drag to pan, right click drag to zoom dynamically.

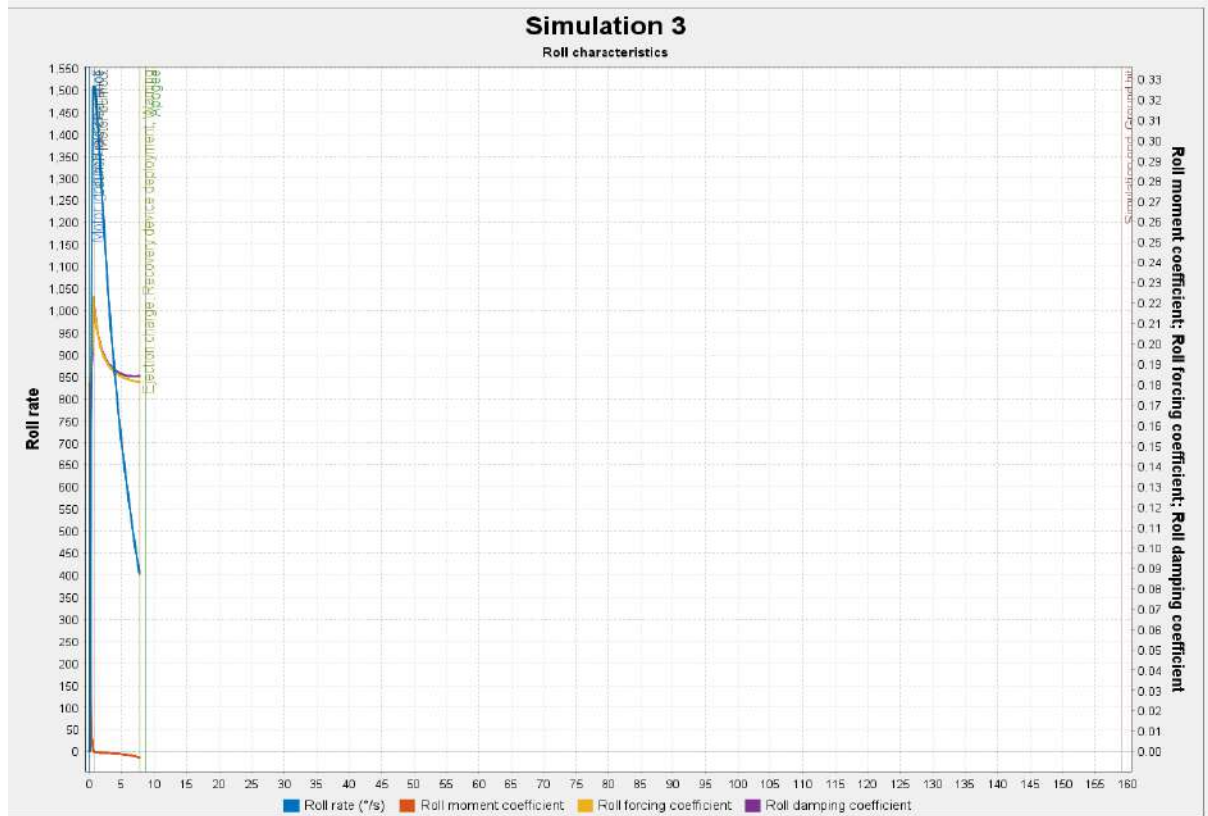


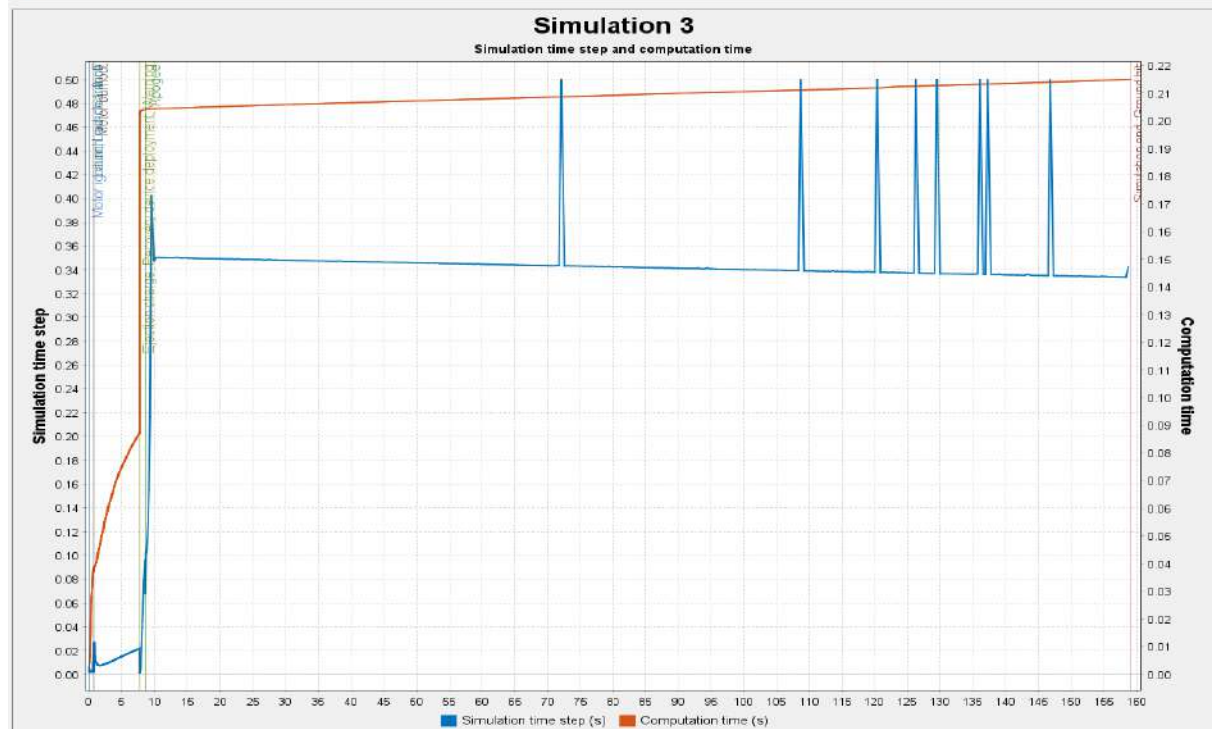
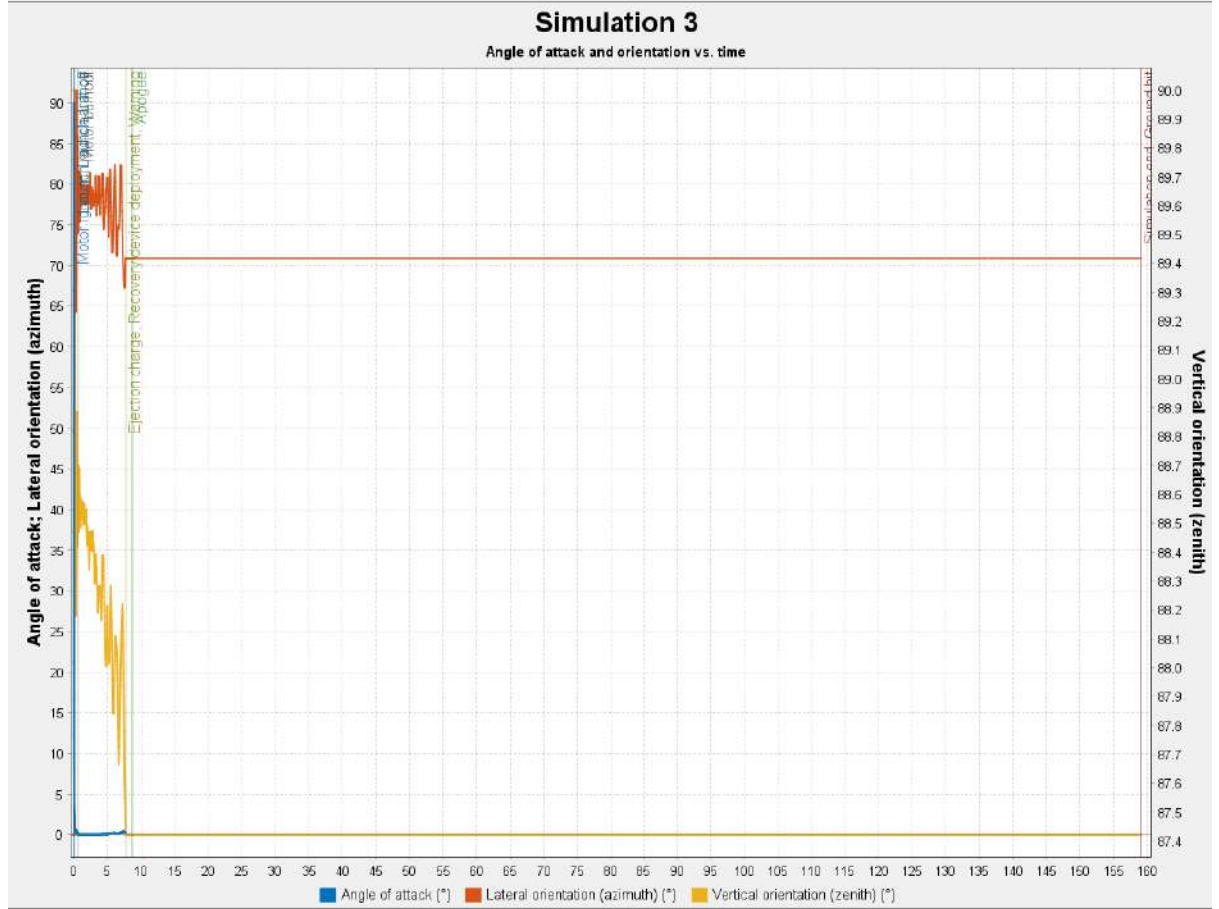


Left click drag to zoom area, mouse wheel to zoom, ctrl-mouse wheel to zoom x axis only, ctrl-left click drag to pan, right click drag to zoom dynamically.

Simulation 3

X





Performance verification was conducted using OpenRocket simulations across all available flight parameters.

The plots collectively confirm compliance with design constraints and training requirements:

- ❖ Apogee exceeds 1000 m*
- ❖ Stability margin remains ~ 1.66 calibers (within 1.0–2.0 range)*
- ❖ Maximum velocity corresponds to Mach 0.798 (< 0.8 , subsonic)*
- ❖ Acceleration profile shows peak $\sim 483 \text{ m/s}^2$*
- ❖ CG and CP locations confirm stable flight*

Additional plots, including thrust curve, descent rate, drag coefficient vs. Mach number, and mass summary, further validate propulsion, aerodynamic behavior, and recovery performance.

Together, these simulations provide a complete verification of the rocket's design, demonstrating both compliance and reliability.

Final Conclusion:-

The rocket design was iteratively refined and validated through OpenRocket simulations to ensure compliance with all training and certification requirements.

Subsystems were carefully selected and dimensioned to balance aerodynamic efficiency, stability, and recovery safety.

Key performance metrics confirm:

- Apogee exceeds 1000 m*
- Stability margin remains ~ 1.66 calibers (within 1.0–2.0 range)*
- Maximum velocity corresponds to Mach 0.798 (< 0.8 , subsonic)*
- Acceleration profile shows peak $\sim 483 \text{ m/s}^2$*
- Recovery descent rate $\sim 6.8 \text{ m/s}$ ($< 10 \text{ m/s}$ requirement)*
- Lift-off mass 702 g ($< 1.5 \text{ kg}$ limit)*

The integration of an ogive nose cone, trapezoidal airfoil fins with through-the-wall tabs, certified H-class motor (H283ST-7), payload with altimeter, and ripstop nylon parachute collectively ensures reliable performance.

Each subsystem contributes to both aerodynamic stability and structural integrity, while the recovery system guarantees safe retrieval.

Thus, the final design is certificate-ready, demonstrating compliance, safety, and reproducibility.

This project highlights a professional workflow where simulation-driven validation precedes documentation, ensuring clarity, authenticity, and reliability in presentation.

